

FOOTBALL PERIODIZATION TO MAXIMISE PERFORMANCE

- Session Design
- The Training Week
- Tapering Strategy
- 102 Practices
- Youth to Pro



Adam Owen PhD

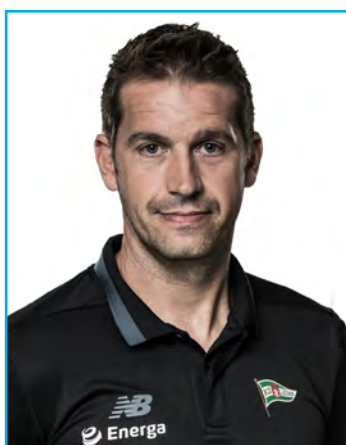


FOOTBALL PERIODIZATION TO MAXIMISE PERFORMANCE

**Session Design | The Training Week |
Tapering Strategy | 102 Practices | Youth to Pro**

Written by

ADAM OWEN Ph.D



UEFA Pro Coaching Licence

Published by



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DR. ADAM OWEN: Coach Profile

UEFA Pro Coaching Licence



Dr. Adam Owen

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Credentials (Coaching and Academic):

- **UEFA Professional Coaching Licence**
- Football Association of Wales (FAW)
- **FA Youth Trainers Award**
- The England Football Association (FA)
- **Doctor of Philosophy (PhD) in Sport Science and Coaching** - Claude Bernard Lyon.1 University, Lyon, France
- **Master of Philosophy Degree (MPhil) in Sport and Exercise Science** - Glyndwr University, Wrexham, Wales, UK
- **BSc (HONS) Degree in Sport and Exercise Science**
- Glyndwr University, Wrexham, Wales, UK

Football Positions:

- **Technical Director**, KKS Lech Poznań, Poland
- **High-Performance Director and Technical Advisor**, Seattle Sounders FC, USA (MLS)
- **High-Performance Director and Assistant Coach**, Hebei China Fortune FC, China
- **Head Coach**, KS Lechia Gdansk, Poland
- **Sport Science and Fitness Coach**, Wales National Team
- **Assistant Manager**, FC Servette, Switzerland
- **High-Performance Director and Assistant Manager**, Sheffield United FC, England
- **Head of Sport Science and Performance**, Rangers FC, Scotland
- **Head of Sport Science and Fitness**, Sheffield Wednesday FC, England
- **Head of Academy Performance and Technical Coach**, Celtic FC, Scotland
- **Academy Head Coach**, Wrexham FC, Wales
- **Player**, Wrexham FC, Wales



Further Roles, Development and Associations:

- **Associate Professorship** at Wrexham Glyndwr University, Wrexham, Wales
- **Associate Researcher (Football Science and Performance)** for Lyon.1 University, Lyon, France
- **UEFA Professional Licence and UEFA A Licence Coach Educator** for the England Football Association
- **UEFA Professional Licence Coach Educator and Coach Developer** for the Finland Football Federation
- **Faculty Member and Lecturer** for the International Soccer Science and Performance Federation (ISSPF) www.ISSPF.com
- **Over 85+ papers published in international peer-reviewed journals including:** Journals of Sport Sciences, International Journal of Sports Medicine, Journal of Strength and Conditioning Research, International Sport Science and Coaching Journal, and many more...
- **Football Consultant Role** with SL Benfica
- **Head of Research and Development** for 5 years at SL Benfica
- **Key Note Speaker** at various international level conferences and congresses



DR. ADAM OWEN: Career Overview of High Performance Expert

Throughout his career, **Dr. Adam Owen** has developed a unique blend of practical coaching experience (**UEFA Pro Coaching Licence holder**) with a very specific and high-level academic profile in **Football Science and Coaching**. Obtaining a Ph.D in the field of Sport Science & Coaching from Lyon 1 University, France, he also holds an associate Professorship role with Glyndwr University, Wrexham in Wales, in addition to continuing as an associate researcher in France and working in the professional game.

Adam's previous **coaching roles** have seen him work across:

- Elite youth level
- Senior level
- UEFA Champions League
- UEFA Europa League
- European club football
- Elite level International football

At the age of 26, Adam was part of the **Rangers FC (Scotland)** management staff who reached the **UEFA Cup Final** in 2008 and remained at the club for seven and half years, gaining valuable experience preparing teams for successful league and cup campaigns in addition to several UEFA Champions League campaigns.

In the summer of 2014, Adam accepted the opportunity to move to **FC Servette (Switzerland)** and experience working abroad at a European club whilst also retaining his role with the **Wales National Team (2009-2018)**.

In 2016, Adam was part of the backroom and coaching staff that reached the **UEFA Euro 2016 Semi-Final** in France, before then becoming **Head Coach in Lechia Gdansk (Poland)**.

Following his experience as a Head Coach at the age of 37 in one of Europe's top leagues, Adam took the opportunity to move to the

Chinese Superleague as a **High-Performance Director**. After a successful 2 year period, just missing out on the Asian Champions League position, Adam joined **MLS Champions Seattle Sounders FC (USA)** as **Technical Advisor** and **High-Performance Coaching Director**. After a 2 year coaching period in the USA, winning the **MLS Western Conference League Title** and reaching the MLS Cup Final, Adam returned to Europe to begin a **Technical Director** role at **Lech Poznań (Poland)**.

Working as an **elite coach educator at UEFA Professional level** within the **England FA** and **Finland FA** to name but a few, Adam has huge experience across many roles within the game:

- Playing
- Coaching
- High Performance Expert
- Manager
- Technical Director

Adam has been able to combine his practical and scientific understanding and experience of the game, in order to mould a specific coaching philosophy outlined in this book.

Adam has published over 90+ football science and coaching articles, book chapters and books, remaining very active in the development of football based research at the elite level, whilst also being a faculty member of the globally recognised International Soccer Science and Performance Federation (www.ISSPF.com), which delivers high-level international online football science and performance courses.

He has been able to utilise previous **European, Asian and North American domestic and inter-continental success to develop a justifiable, research based coaching method in order to maximise individual and group performance within elite professional football.**

INTRODUCTION:

Training Sessions for Peak Performance

Throughout the last decade or so, the training and match play demands imposed upon elite level football players have grown enormously in order to meet the high conditioning requirements of increased fixtures, travel and subsequent domestic, continental and international based competitions.

In order for players to cope with increased repetitive match related demands at near maximal performance levels during competition, the desire and need of coaches to understand, control, analyse, and eventually manipulate training sessions has increased.

As the modern game continues to rapidly evolve across various aspects of the game, a greater range of tactical understanding, situations and subsequent playing demands require players to have greater physiological capacity, in addition to an enhanced psychological understanding in and out of possession compared with previous decades (Križaj et al., 2019).

It is well-documented that **football is now played at a quicker pace** with the ball being in play ~12 minutes more than early reports in the 1990's, coupled with the fact **significantly more high-intensity actions and movements are performed during the game** (Mohr et al., 2003).

One such report (Barnes et al., 2014) in this area has shown how at the very elite level:

1. **High-intensity running distance has increased by 30% since the early 2000's**
2. **Sprint demands have nearly doubled since the early 2000's**

To positively influence performance levels, the most efficient methods to improve individuals or teams to the best of their ability must be used. To achieve this, the **training structure must be planned and prepared adequately through justifiable methodologies,** to therefore **maximise levels of performance.**

Coaches must ensure:

- **Players progress in-line with contemporary and innovative changes within the game**
- **Drive players to the next level from a technical, tactical and physical perspective (primary aim of a coach)**
- **Maximise the training time and efficiency of the coaching process**
- **Maximise the complex nature of football development in respect of technical and tactical abilities**

Psychosocial and cognitive skills place a huge multi-dimensional demand in the planning and training session design phase. Throughout the season, technical and performance practitioners consistently search for best practice in order to justify:

1. **Training objectives**
2. **Apply weekly tapering strategies**
3. **Apply specific training programs to achieve specific outcomes** (incorporating all the vital aspects of performance development)

When trying to apply a holistic coaching process (where every aspect of a player's development is taken into account), it is vitally important to select a coaching process or approach with the target of providing the best decisions for performance development.

When discussing a holistic coaching approach within the game and maximising player performance as a coach, it is necessary to continually expand the understanding of the game using research as a way of making better decisions, justifications and evolution. Not only can 'football science' research assist in making better coach-related decisions, but it can also lead to the development of new theory or methodologies to drive the game forward.

Various research into training methodology, nutrition, psychology, as well as the testing and monitoring of players, has recently led to managers, head coaches, performance coaches and technical support staff being able to produce much greater efficiency.

Irrespective of the level of players being coached within the game, all coaches face the very relevant question based on the various challenges needed to gain results:

- What do I need to do in order to get the best possible performance from my group of players?
- What can I change or improve to maximise players collectively as individuals and subsequently as a team?
- What gives me the best possible chance of succeeding as a coach?

Addressing these questions is something all individuals within the game want to achieve. In order to attempt to discuss this within a football environment, one of the most debated and hot topics include detailed discussions around:

- **Training Periodization**
- or **Tapering Strategies**
- and **Training Methodology**

The intentions behind this book are as follows:

- Utilise **applied football science and performance coaching** research, in collaboration with a football specific methodology **which can be used across a range of coaching levels.**
- Engage coaches with a thirst to evolve or understand more.
- This book provides a **unique blend of modern football coaching practices, with an innovative coaching theory and methodology.**
- Expose coaches to various topics such as the **development of a coaching methodology, understanding what is meant by 'game model'** in a football context, and teach a **game based training approach.**
- Show **how to build training sessions.**
- Teach to **practically apply the training content.**
- Consider and demonstrate **how each training aspect fits into the flow of the session and training week.**
- Show the actual demands imposed on players within the training and competitive environment (so you can adequately prepare the coaching process).
- **Training Load Management** - an overview of what coaches can do to ensure their players remain in the best state to **optimise performance, whilst reducing injury risk at the same time.**
- Gaining the balance of fitness vs. freshness to perform on a match-day.

Benefits of the "Football Periodization to Maximise Performance" Methodology

Key point

If the daily training content is poorly planned or managed through an insufficient methodological approach, poor performance occurs with the players insufficiently conditioned and significantly higher rates of injury.

What changes have there been in the last decade?

Individuals tasked with the development of football players have seen the understanding, progression and implementation of strength and conditioning, speed development and high-intensity football specific endurance training increase exponentially over the last decade. Furthermore, that has happened directly in accordance with growth in the players' tactical understanding of different systems of play.

What are the benefits?

- Enhancing the coaches' knowledge across a range of coaching and high-performance topics
- Maximising the use of specific training games and practices in the training week (microcycle)
- Understanding the benefits of tapering strategies = players arrive in optimal condition for competition!

The overall picture of the coaching process

Maximising the physical profile and status of players is only a part of the performance target, as from a coach's perspective, building an integrated training process to a level where the physical, technical and tactical outcomes seamlessly fit into the development of the player or team, is fundamental to obtain performance progression within the coaching process.

FOREWORD by Steve McClaren



- **FIFA Technical Advisor**
- **England National Team Head Coach**
- **Assistant Manager of Manchester United** (1 x Champions League, 3 x Premier League, 1 x FA Cup including 1999 Treble)
- **Middlesbrough FC Manager**
- **FC Twente Manager**
- **VfL Wolfsburg Manager**
- **Newcastle United FC Manager**
- **Derby County FC Manager**

"Throughout my career in the game as a coach, assistant manager or manager, one of the greatest shifts within the professional side of the game has come through the integration of performance science and training methodology.

Improving the performance of players and trying to gain a competitive advantage is something every coach desires.

The developments and evolution of **modern coaching education has been fundamental**, and becomes more evident amongst the elite teams when it comes to the **tactical and physical preparation, training methodology** and subsequent **match performance**.

Understanding the tactical demands of the game is significant to all individuals wanting to improve their work in the game. However, how we train or

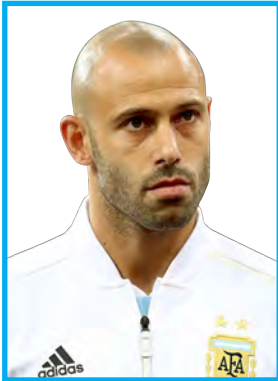
coach players to perform in the best possible condition should be the key target of all coaches.

Technical coaching staff, as well as performance specialists, will enjoy the applied coaching focus, methodology and topics discussed within this book.

Maximising the key coaching link between the technical, tactical and physical details of the game, **this book perfectly blends the research, coaching education and current trends within the game with practical integrated coaching details.**

As a result, this book provides a **great coaching resource for all individuals wanting to develop their knowledge of the game.**"

Player and Content References



Javier Mascherano

- Current Argentina Under 20 National Team Manager
- Former FC Barcelona, Liverpool FC, and Argentina National Team player

"Adam and I started working together when he arrived in China. He is a great professional and person who shows a high level of passion and quality within his working methodology in football."

I am grateful to him for the time we were working together, as he was a great help to me both professionally and personally. We remain in contact, and I hope we will get the chance to work together again in the future."



Gareth Bale

- Real Madrid and Wales National Team player

"Adam is someone I have always trusted and whose opinion and advice I trust. We remained in constant communication between international duties and always had an excellent personal and professional relationship."

Having worked for so long together at the elite international level of the game, it shows his professional quality and worth.

Adam is someone who has helped me through my career and is someone I always enjoyed working with on a day-to-day basis."



Aaron Ramsey

- Juventus FC and Wales National Team player

"After knowing and working with Adam for many years internationally, I consider him to be a top professional in his area of expertise, and someone with whom I have been able to improve different areas of my game, due to his methodology and knowledge."



Nicolás Lodeiro

- Former Ajax, Boca Juniors and Uruguay player
- Current Seattle Sounders FC and MLS All-Star player

"Adam came to the MLS and implemented a first class training structure which was a big part of our club's success.

Adam's coaching approach was very detailed and one that I enjoyed working within due to it being very football specific and high-intensity. His integrated coaching methodology, training sessions and preparation helped me significantly through his time at the club."



Dr. Andreas Schlumberger, PhD

- Current Head of Recovery and Performance at Liverpool FC
- Former Head of Rehabilitation and Prevention at FC Bayern Munich

"Adam provides an outstanding mixture of applied research, practical application and tremendous experience in the world of high-performance football.

With this very unique approach, he will continue to have a big impact on criteria-based and functional work in all applied areas of performance development and coaching in football."



Dr. Jorge Candel, MD

- Currently Head of Medicine, Clinica Tecma, Spain
- Former Head of Medical Department and Club Doctor for CF Valencia

"Having worked with Adam for a number of years and seen his approach to both coaching and performance science, he is someone I feel will remain at the top of his profession for many years to come.

The contents of this book are of great interest to all individuals interested in increasing their knowledge of the game from a practical and academic perspective."

CHAPTER 1

Coaching Methodology to Maximise Performance

Coaching Methodology to Maximise Performance

Coaches, performance experts and physical specialists involved with the preparation of football players constantly research methods or ways to analyse and examine the training loads, **aiming to maximise performance**.

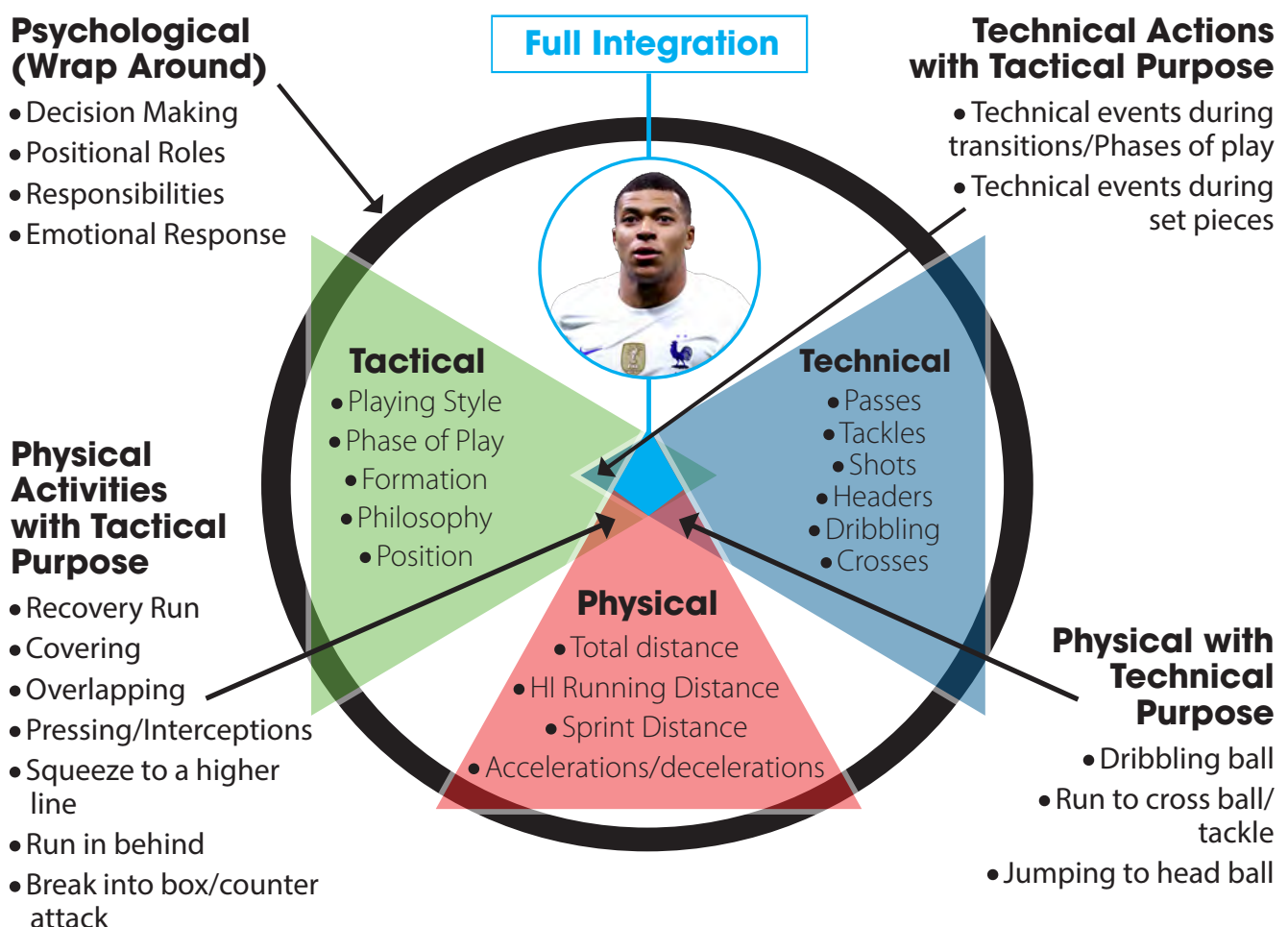
Maximising performance is determined by a great number of factors within football. However, at the elite level of the game, these are the **3 key starting points**:

1. Balancing fitness vs. freshness
2. Encouraging physical robustness
3. Mental resilience

Exposing players to key physical, psychological, technical and tactical demands across the training week is the fundamental aspect of player preparation.

This involves not only **assessing training periodization and tapering strategies**, but also recognising and justifying the distinct requirements and effects imposed on players through various training scenarios (Owen et al., 2014), the various positional demands, and the conditioning needs of the individual within competitive match-play (Owen et al., 2016).

Game Demands: Intersections and Fully Integrated Training Approach - Adapted from Bradley, et al., (2010)



Training Load, Adaptation, and Training Week (Microcycle) Tapering

Research surrounding the training session design and microcycle structuring has provided evidence of how players in different positions reveal varied outcomes in terms of key physical metrics monitored:

- Total Distance Covered (TDC)
- High Speed Running (HSR)
- Sprinting

It is **vitaly important to understand how training sessions or coaching instructions influence these metrics, and this should be taken under consideration when programming the structure of the weekly microcycle** (Owen et al., 2017; Malone et al., 2018; Martin-Garcia et al., 2018).

Understanding that the **internal load responses of training is what causes**

significant adaptations is an integral foundation for any coach at any level of the game, based on the impact of fatigue, recovery and the physiological adaptation process that occurs post-training.

Adaptation: The process of the body getting used to a particular training program through repeated exposure.

The microcycle and training structure is dictated by the physical recovery status and the conditioning requirements of the players in relation to the upcoming match. **Microcycle tapering strategies enable players to be in a better prepared physical, psychological and physiological state for competitive matches** (van Winkle et al., 2014; Malone et al., 2016; Fessi et al., 2016; Owen et al., 2017).

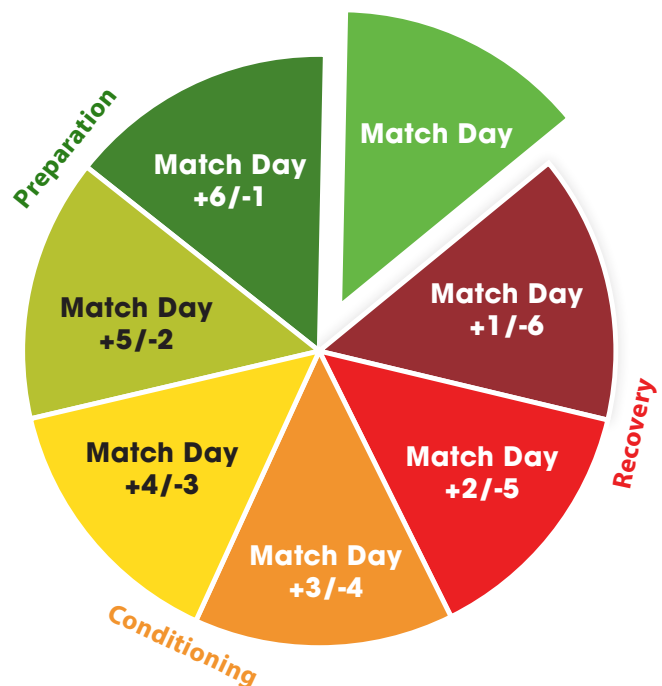
Load Management of the Weekly Microcycle (Training Week)

- Focus post-match = **Recovery**
- Research = **48hrs post-match key** to reduce physical, psychological, and biochemical fatigue
- Impact of **daily session** on the next day
- Cyclical basis and raised importance of **session design**



Correct loading strategy through the week employed to:

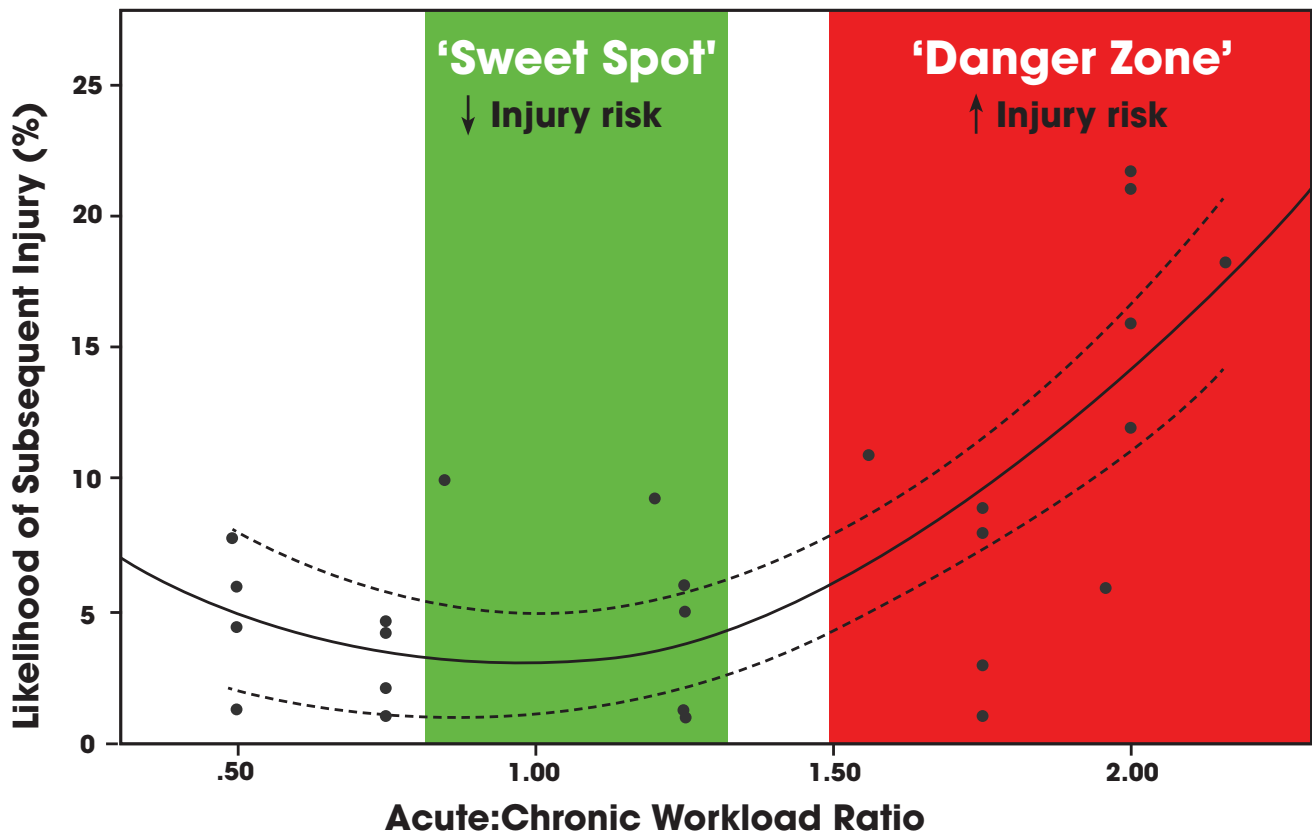
1. Cause adaptation
2. Recovery for match day



CHAPTER 3

Training Load Management in Football

Likelihood of Subsequent Injury vs. Workload Ratio



Fatigue, Recovery & Endurance

Adding significant training load to underprepared players within the training process or cycle inevitably leads to increased accumulation of fatigue. Based on the specific recovery needs generated by the fatigue, certain individuals may be affected in different ways due to several factors:

- Severity and type of exercises inducing the fatigue
- Training age
- Physical fitness and conditioning levels

Early research performed by Helgerud et al., (2001) using elite football players showed highly individual trained physical capacities are associated with significant performance benefits, such as **increased aerobic capacity leading to greater total distance covered (TDC), number of sprints, improved recovery and greater ball-involvement.**

In addition, recent research in this area have also described how relationships exist between endurance levels and **Total Distance Covered (TDC)** in competitive match-play, which also promoted an increased **Maximal Aerobic Speed** (Swaby et al., 2016).

Adding more credibility to these reports, it is suggested that aerobic endurance capacity is an important consideration with high maximal values correlating with match-play work rate. This results in significant benefits which aid the recovery periods between high intensity playing periods.

Furthermore, developing the training capacity of players offers a protective effect from subsequent injury risk, rather than limiting the training response in fear of injury (Gabbett, 2016).

High Speed Running (HSR) and Sprinting (Maximal Velocity)

The investigation by Malone et al., (2018) into athletic capability of footballers found players performing significantly higher during the **30-15 Intermittent Fitness Test (IFT)** [30-second shuttle runs interspersed with 15-second walking recovery periods] tolerated greater volumes of **High Speed Running (HSR)** and **Sprint Distance** when compared with players revealing a reduced aerobic fitness. Having highlighted the importance of developing the players aerobic endurance capacity within football through practices/drills and sessions causing a **cardiovascular overload (small sided games and high-intensity resistance training drills)**, the significant role sprinting and HSR exercises play in reducing injury risk and developing all round-robust players should be considered.

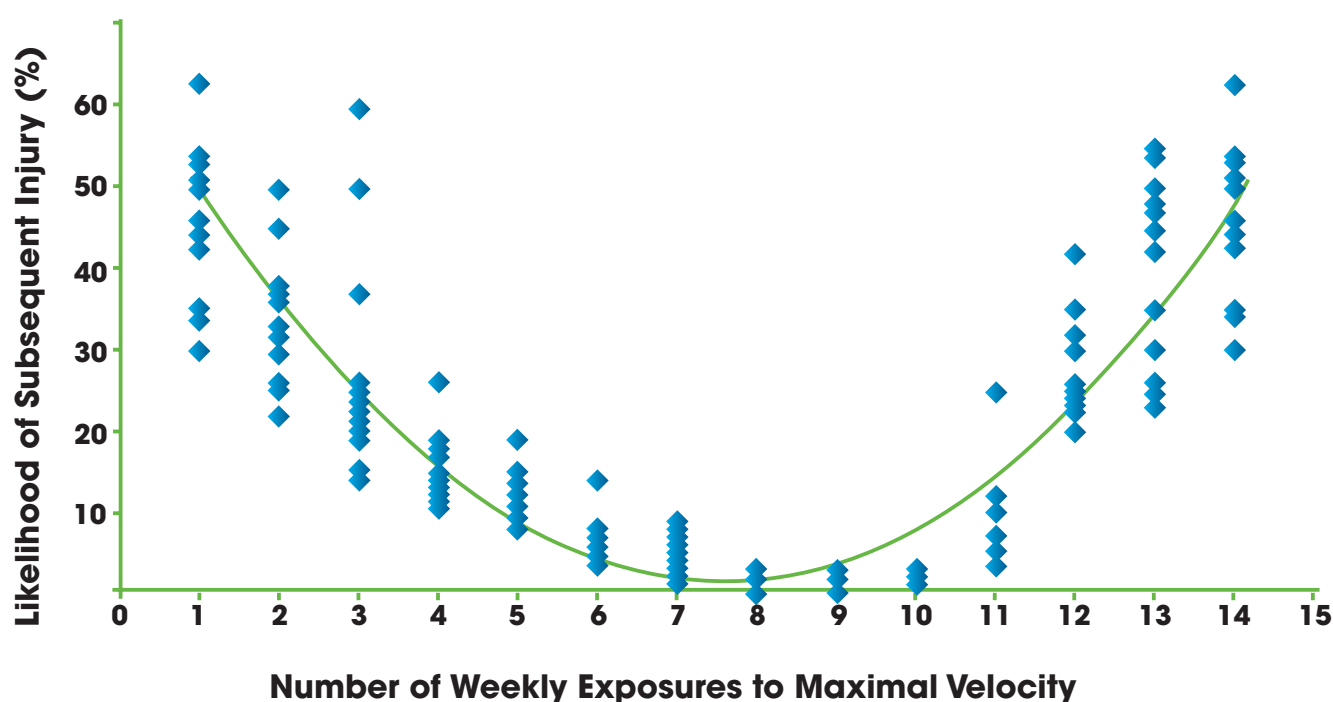
An earlier study by Malone et al., (2016) examined the relationship between sprint exposures and injury risk in Gaelic footballers. The findings reported how **players who achieved >95% of their peak**

velocity had a reduced injury risk when compared to players exposed to lower relative velocity. Further detailed analysis of the data revealed how **players who performed excessive loads of peak velocity actions were at increased risk of injury, highlighting a “U” shaped relationship between maximum velocity exposures and injury risk (please see the graph below).**

Players performing between 6 to 10 sprints >95% Maximum Velocity (Vmax) capacity had a lower injury risk versus players performing less than 5, or more than 11 sprints. Results of this study therefore provide **evidence of a possible ideal sprint stimulus as a method to minimise injury risk in team sports.**

Conversely, when exposed to the same amount of sprinting, players with higher chronic loads had a lower injury risk. Exposing players to rapid increases in HSR and sprint distances increased the odds of injury. However, higher chronic training loads and better intermittent aerobic fitness off-set lower limb injury risk associated with these running distances (Malone et al., 2018).

Association Between Total Weekly Maximal Velocity Exposures and Likelihood of Injury - Taken from Malone et al., (2016)



CHAPTER 4

Developing a Game Based Training Approach

PHYSICAL AND TECHNICAL COMPARISONS BETWEEN VARIOUS SIDED GAMES WITHIN PROFESSIONAL FOOTBALL

Reference: Owen, Wong, Paul & Dellal, IJSM 2014

Designed by @YLMsPortScience

10 Elite Players Studied (3 x 5 min each game type)

SMALL (SSG)

1v1-4v4 +GKs

vs

MEDIUM (MSG)

5v5-7v7 +GKs

vs

LARGE (LSG)

8v8-10v10 +GKs



RESULTS

1. Small sided games (SSGs) induce a faster speed of play when compared to MSGs (+39%) and LSGs (+26%)
2. ... but induce less repeated high-intensity efforts (0.88 vs 4.40 m), high-intensity running (7 vs 39m) and sprint distance (0 vs 11m) when compared to large sided games (LSGs)
3. SSGs have more passing, receiving, dribbling, and shooting compared to medium sided games (MSGs) and LSGs
4. MSGs have more passing and shooting than LSGs

Conducting the correct type of sided game at specific times of the training week may enable you to optimally prepare players physically, technically and tactically, thus increasing the efficiency of your training sessions and the weekly schedule

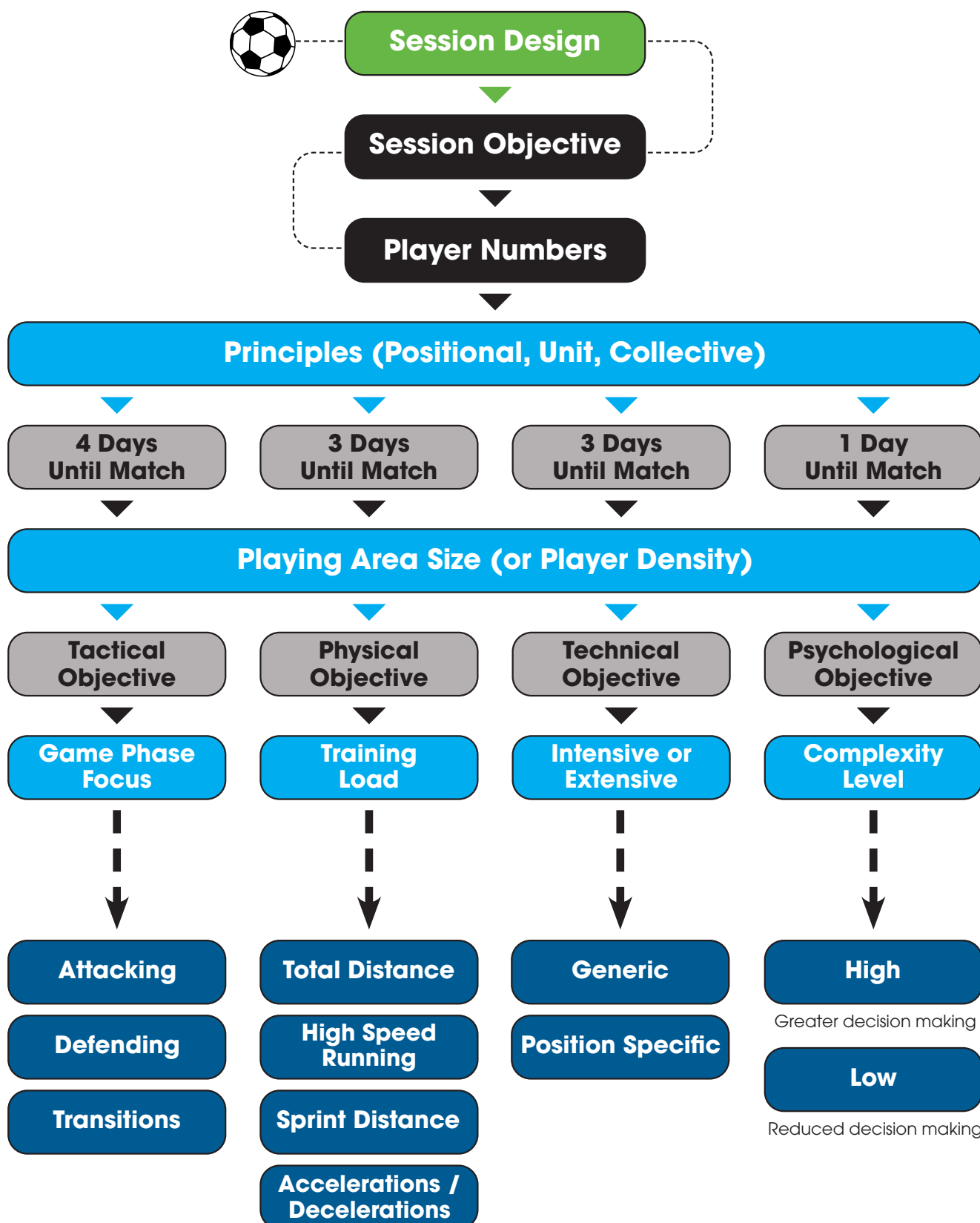


Images provided by PresenterMedia

CHAPTER 5

Training Session Design

Practice Design Considerations to Optimise Coaching Outcomes



SKILLS PERIODIZATION

Reference: Mujika et al. IJSP 2018

Designed by @YLMSPortScience

SPECIFICITY

Specificity of the practice/s is an essential element of a periodized plan (specific time blocks)

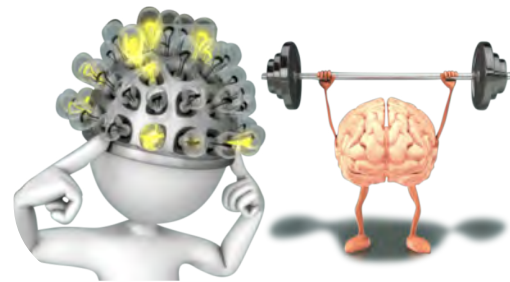
The complexity of specificity could be manipulated through variations in the constraints applied e.g. amount of defensive pressure, time pressure, but some form of specificity is always present.



OVERLOAD

Cognitive effort usually indicates that the performer is required to actively engage in skill practice

Opportunities to unload the degree of cognitive effort is also critical, particularly when considered in connection (parallel) with the physical training load.



PROGRESSION

Progression in the complexity of the practiced skill and its interaction with the amount of repetitions (frequency) completed



Complexity and frequency can be manipulated (and recorded) to develop an overall load which brings about an optimal challenge point.

PREVENT REVERSIBILITY

Reversibility highlights the importance of systematically recording skill performance to determine the degree of learning achieved



Identifying how long a skill can be left without practice before reversibility effects appear is useful in periodized high performance programs often overcrowded with competing practice needs and limited practice time.

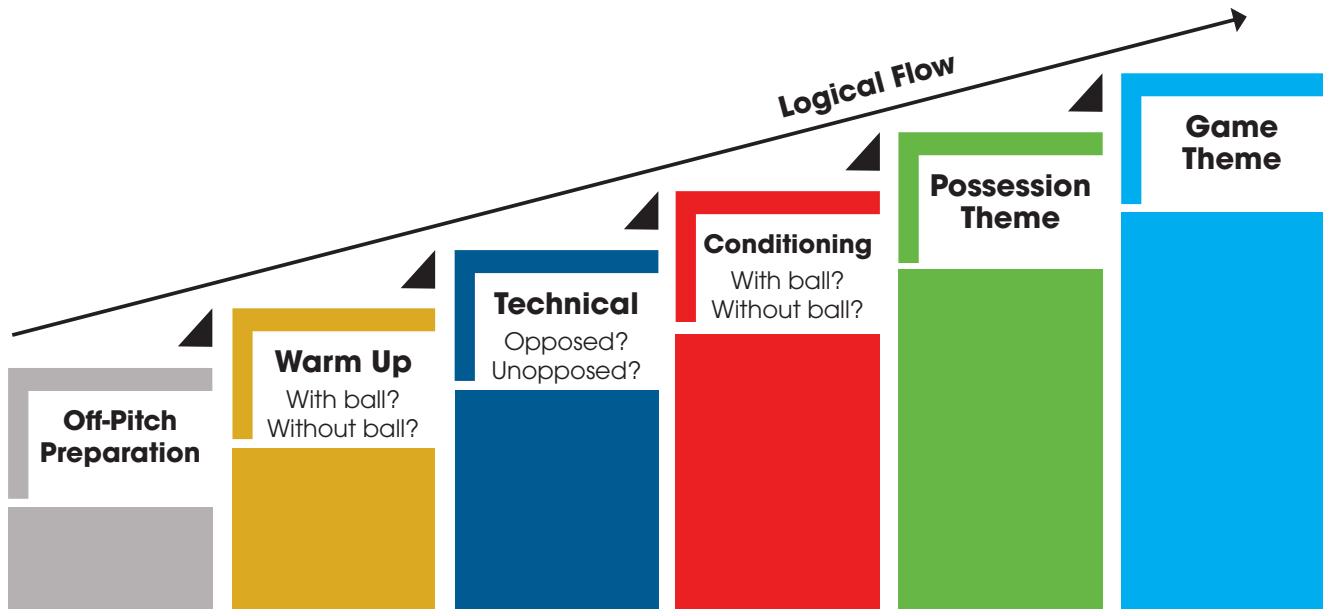
PREVENT BOREDOM

Increased practice variability is useful for reducing the likelihood of boredom. Higher practice variability is associated with suppressed practice performance but superior transfer performance and increased cognitive effort. These interactive features need to be understood for skill development periodization.

Boredom due to monotony is detrimental to any skill development

Images provided by PresenterMedia

Training Session Flow



- Ensure logical flow through session – physical, technical, and tactical
- Intelligent and efficient coaching – maintain focus and intensity
- Be concise and direct with coaching points – maintain clarity
- Use natural breaks to coach and get points across
- Don't break the rhythm and reduce intensity
- Sessions should flow!

Warm-up

- Don't just use as a time filler!
- Physically prepare for session demands
- Psychological preparation (tactical)

Technical

- Continue flow of the session
- Gradually increase intensity and demand from warm-up phase
- Influence technical, tactical, psychological and physical outcomes
- Expansion or limitation of area size depending on physical requirements

Conditioning

- Overload the physical focus of session
- Induce key physical stimulus of session
- Ensure opportunity for players to develop specific physical qualities
- Prepare players fully for the upcoming intensity demands

Possession

- Continue flow and demand of the session
- Ensure coaching within the phase is concise and intelligent
- Ensure match intensity is attained to achieve competition transfer and realism
- Coaching within natural recovery breaks is key to minimise influencing the intensity and direction of the session

Planning the Training Week with Tapering Strategy to Maximise Performance

To assist in the practical application of the content, the flow of the training session is vitally important to increase the intensity, application and engagement of the players. This can be done through administering the content of the session through a logical flow, focussing on some simple key coaching points.

With good preparation, both tactically and physically, the players' roles and responsibilities can be understood further and lead to improved performances.

The next chapter 6 shows you how to structure the training week (periodized microcycle) with a tapered strategy so your players can reach their peak performance. Training week plans for Professional, Semi-professional, Youth Academy, and Grassroots (Youth) football are all included.

The following chapter 7 is "**Periodized Practices to Maximise Performance**" which shows you all the different types of practices organised into their different sections:

- » **Recovery**
- » **Warm-ups** (Resistance, Speed Endurance, Reaction Speed)
- » **Technical** (Intensive, Extensive)
- » **Conditioning** (Resistance, Speed Endurance, Reaction Speed)
- » **Possession** (Small, Medium, Large)
- » **Games** (Small, Medium, Large)

You can then drop these practice types into the applicable day and session on the training week session plans.



CHAPTER 6

Planning the Training Week with Tapering Strategy to Maximise Performance

Practical Coaching Model to Build the Training Week (Microcycle)

Using the training sessions within this book, the tables to follow provide a framework for coaches of various coaching levels or categories to utilise as a guide in order to structure their training week. Following this specific training methodology, it is possible to implement an integrated training concept. Being able to understand the physical, technical and tactical outcomes of the session are key to maximising the coaching time with the players involved.

Selecting from the various categorised practices in this book in the correct order provides assistance with the fundamentally important session design phase.

Furthermore, based on the understanding of the physical outcomes of each practice, coaches will be able to enhance their knowledge of how **the session design phase can be tailored to meet the session objectives from a physical, technical and tactical perspective.**

Key Point: Selection of practices in the book will generate a better understanding for coaches of the physical demands imposed by individual practices and accumulative total sessions over a period of time.

It should be noted that the physical data provided in these practices and sessions within this book has been generated from elite professional players, so it is suggested that practice durations, repetitions and area sizes are adapted to best suit the age groups being coached.

The data values give the readers an understanding of the demands imposed on players at the level assessed. The physical output metrics are for coaches to understand how different practices can influence different physical loads.

The 5 different training week examples (microcycles) outlined on the following pages are as follows:

- 1. Professional Microcycle**
(4 Training Sessions per week + Match + Compensatory Session)
- 2. Semi-professional Microcycle**
(3 Training Sessions per week + Match)
- 3. Youth Academy Microcycle**
(2 Training Sessions per week + Match)
- 4. Grassroots (Youth) Microcycle 1 - Small Sided Game Focus**
(1 Training Session per week, which alternates with Grassroots Microcycle 2)
- 5. Grassroots (Youth) Microcycle 2 - Large Sided Game Focus**
(1 Training Session per week, which alternates with Grassroots Microcycle 1)

NOTE: In the next chapter of the book, we have presented the practices in the format of the **Professional 4 Training Sessions + Match training week**, but you can adapt it and select the practices required for your relevant 3, 2 or 1 session training week.

THE TRAINING WEEK:

Professional Microcycle

4 Training Sessions per Week + Match + Compensatory Session

DAY OF THE WEEK		MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
Post-Game + / Pre-Game -		MD +2/-5	MD +3/-4	MD +4/-3	MD +5/-2	MD +6/-1	Match	MD +1/-6
Game Focus		Recovery	Intensive	Extensive	Balanced	Intensive	Extensive	Non-Starters
Tactical Focus		Evaluate	Defending	Attacking	Balanced	Review	Execute	-
PHYSICAL FOCUS		RECOVERY	RESISTANCE	SPEED END.	REACTION SPEED	ACTIVATION	MATCH	COMPENSATORY
Warm-up	Recovery							
	Resistance							
	Speed Endurance							
	Reaction Speed							
Technical	Intensive							
	Extensive							
Conditioning	Resistance							
	Speed Endurance							
	Reaction Speed							
Possession	Small Sided							
	Medium Sided							
	Large Sided							
Game	Small Sided							
	Medium Sided							
	Large Sided							

This shows how the colour-coded practices to follow in the book can form the training week based on periodized training sessions to optimise performance.

This example shows a specific methodology of work across the microcycle for professional or full-time training teams. Obviously, this is a template to work from because every coach has their own individual way of working when it comes to session flow, tactical focus, etc. The compensatory based training content is left blank for the coach to fill in.

Training Session Format Example for Professional Training Week (Microcycle)

SUNDAY/MONDAY - 1/2 Days Until Match = Recovery

TUESDAY (70-75 min) - 4 Days Until Match (MD +3/-4)
Positional Principle Training and Resistance:

1. Resistance Warm-up (10-12 min)
 2. Intensive Technical Practice (10-15 min)
 3. Resistance Conditioning Practice (10-20 min)
 4. Small Sided Possession (10-12 min)
 5. Small Sided Game (10-25 min)
-

WEDNESDAY (85-95 min) - 3 Days Until Match (MD +4/-3)
Collective Team Principle Training and Speed Endurance:

1. Speed Endurance Warm-up (10-12 min)
 2. Extensive Technical Practice (12-15 min)
 3. Speed Endurance Conditioning Practice (5-15 min)
 4. Large Sided Possession (10-15 min)
 5. Large Sided Game in Large Area (10-50 min)
-

THURSDAY (60-70 min) - 2 Days Until Match (MD +5/-2)
Unit Principle Training and Reaction Speed Development:

1. Reaction Speed Warm-up (5-7 min)
 2. Intensive Technical Practice (10-15 min)
 3. Reaction Speed Conditioning Practice (5-15 min)
 4. Medium Sided Possession (6-15 min)
 5. Medium Sided Game (10-25 min)
-

FRIDAY (45-60 min) - 1 Day Until Match (MD +6/-1)
Pre-Match Activation Training Day:

1. Resistance Warm-up (10-12 min)
2. Reaction Speed Conditioning Practice (5-15 min)
3. Large Sided Game in Small/Medium Area (10-50 min)

THE TRAINING WEEK:

Grassroots (Youth) Microcycle 1

SSG/Resistance Focus - Alternates with Grassroots (Youth) Microcycle 2

DAY OF THE WEEK		MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
Post-Game + / Pre-Game -		MD +2/-5	MD +3/-4	MD +4/-3	MD +5/-2	MD +6/-1	Match	MD +1/-6
Game Focus		Recovery	Recovery	Intensive	Recovery	Recovery	Extensive	Recovery
Tactical Focus		Free Evening	Free Evening	Attacking	Free Evening	Free Evening	Execute	Free Evening
PHYSICAL FOCUS		RECOVERY	RECOVERY	RESISTANCE	RECOVERY	RECOVERY	MATCH	RECOVERY
Warm-up	Recovery							
	Resistance							
	Speed Endurance							
	Reaction Speed							
Technical	Intensive							
	Extensive							
Conditioning	Resistance							
	Speed Endurance							
	Reaction Speed							
Possession	Small Sided							
	Medium Sided							
	Large Sided							
Game	Small Sided							
	Medium Sided							
	Large Sided							

This shows how the colour-coded practices to follow in the book can form the training week based on periodized training sessions to optimise performance.

Grassroots (Youth) Microcycle 1 has a small sided game focus and alternates with Grassroots (Youth) 2 - [see next page](#), which has a large sided game focus

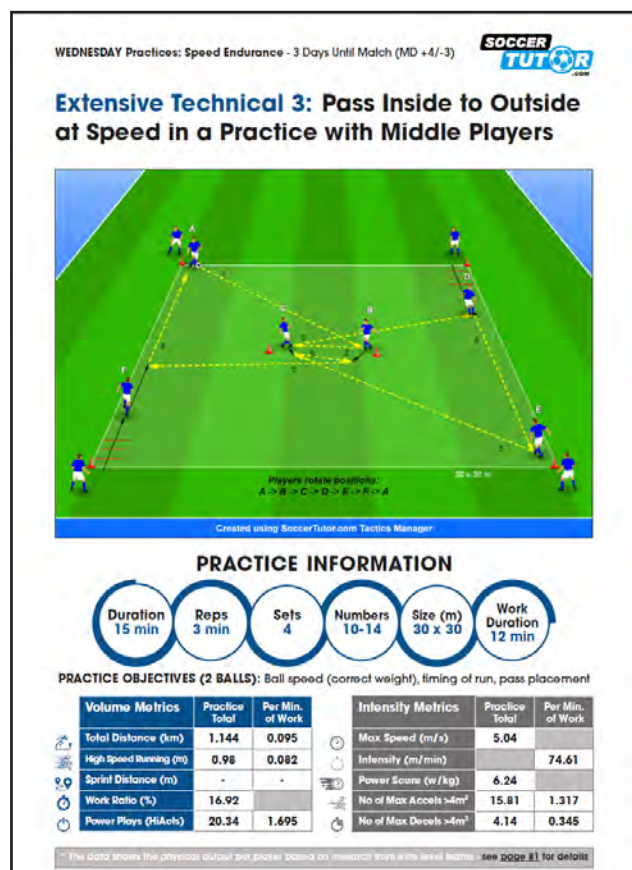
Obviously, this is a template to work from because every coach has their own individual way of working when it comes to session flow, tactical focus, etc.

CHAPTER 7

Periodized Practices to Maximise Performance

NOTE: In this chapter, we have presented the practices in the format of the **Professional 4 Training Sessions + Match training week (see pages 74-75)**, but you can adapt it and select the practices required for your relevant 3, 2 or 1 session training week (please see relevant tables in the previous chapter).

The Benefit of the Data for Each Practice (Volume and Intensity Metrics)



Where does the data come from?

The data is collected from the elite professional level using GPS tracking equipment.

How can coaches of all levels learn from this data?

In my role as a **coach educator** for Level 2, UEFA B, UEFA A and UEFA Pro Licence, all of the coaches have been extremely excited to have access to the data included in this book.

The information provides a method for all coaches tasked with the performance development of players. The data provides key information from the professional level (average per player), in order to educate the differences between different types of practices and their specific physical loads for coaching at all levels.

It is extremely important to understand how different practices produce specific outcomes and physical development stresses to the players within those sessions. This is key for **tapering sessions and weekly training plans to reduce the risk of injury and for optimising the specific coaching focus from a physical conditioning perspective.**

Why are the volume and intensity metrics useful information to have for each practice?

The values give the readers an understanding of the demands imposed on players. It is extremely important to understand how different practices produce different physical outcomes and stresses. The data figures show you this e.g. Total Distance Covered (m), Work Ratio (%), Sprint Distance (m), etc.

How can this data help me optimise my training plan to produce maximum (peak) level performances?

Selection of practices in the book will generate a better understanding of the physical demands imposed by individual practices and accumulative total sessions over a period of time.

Understanding the physical demands on specific training practices and sessions is a vitally

important aspect of the modern coaching strategy to maximise all aspects of training.

The main aim of any training load management process is to provide the individuals involved in the football development of players to positively evaluate and interpret the data they have available to them.

The main point is that all coaches need to now be aware of this information and data, so they can produce the best possible results.

If the same mistakes keep occurring without monitoring relationships between training load and the game, then we may just be guessing, potentially regressing the players development, and providing poor quality training to our players...

KEY TERMS

VOLUME METRICS FOR ALL PRACTICES



Total Distance (km)

Total distance provides a full representation of volume of exercise (walking, running, sprinting, jogging) and is a very simple way of assessing individual effort within the practice or game.



High Speed Running (m)

HSR is distance travelled above speeds of 5.5 metres per second. Greater amounts of high speed running signify a high level of the game or football fitness, however different playing surface areas influence this metric assessed within training. Players will commonly cover HSR distances above 1000-1500m depending on position, maximum speed and fitness levels.



Sprint Distance (m)

Sprint distance is calculated as total distance covered above 7 metres per second. The capacity of players to achieve this is generally higher amongst elite professional players. The playing area significantly influences this value as small sided, reduced areas lower the sprint opportunities. Greater amounts of sprinting signify a high level of game or sport fitness.



Work Ratio (%)

This is defined as the percentage of time the player was performing work or movements. The work is defined as walking or running at speeds higher than 1.5 metres per second (slow to moderate walk for most people). Work ratio in general can be associated with the amount of time a player is working compared to resting during a training session, or game phase.



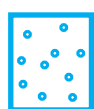
High Metabolic Load Distance (HMLD)

HMLD measures the total amount of high speed running performed, coupled with the total distance of accelerations and decelerations throughout a session.



Power Plays (HiActs)

This is defined as a significant action (such as acceleration or high speed running event) in which the power output performed by the player was above 20 watts per kg of body weight. Counting power plays gives you an indication of the number of intense actions the players performed in the practice or game. These are obviously far more physically demanding.



Player Density (m²)

The quantity per unit of playing space per player, which is calculated as follows:
Length of pitch x width of pitch, divided (÷) by the number of players on the pitch.

INTENSITY METRICS FOR ALL PRACTICES



Max Speed (m/s)

Max speed is defined in this book as the fastest maximum speed achieved and sustained for at least half a second. They are represented as metres per second (m/s). For most players, 8.5m/s can be considered quick, but when we compare to elite level sprinters such as world record holder and legendary athlete Usain Bolt, he achieved regular speeds ~11.1m/s.



Intensity (m/min)

Distance per minute provides the coaches and players with an overall representation of how hard, or how intense they have worked. Professional players within competitive match play can achieve between 112–135 m/min, however these results are different depending on the tactical strategy played, and positional differences of the individual players.



Power Score (w/kg)

Calculated as watts (w) divided by your body weight in kilograms (kg) = w/kg. This provides an insight into the power output used per kg of the player's weight and is used to gauge the intensity of training practices with high work-rates within a small area e.g. Small sided games. Practices are considered intense when the power score is above 10 w/kg. Amongst amateur football players, values of 7–8 w/kg are normal.



Number of Max Accelerations >4m²

Number of accelerations performed in the practice/drill or session that are greater than 4 metres per second². These are the higher, more explosive accelerations that demand significant energy, strength and power.



Number of Max Decelerations >4m²

Number of decelerations performed in the practice/drill or session that are greater than 4 metres per second². These are the higher, more explosive decelerations that demand significant energy, strength and power.



Max Acceleration Distance (m)

Total distance accumulated within the session through accelerations performed greater than 4 metres per second². These are the higher, more explosive acceleration distances.



Max Deceleration Distance (m)

Total distance accumulated within the session through decelerations performed greater than 4 metres per second². These are the higher, more explosive deceleration distances.

WEDNESDAY TRAINING DAY PRACTICES: **SPEED ENDURANCE**

3 DAYS UNTIL MATCH (MD +4/-3)

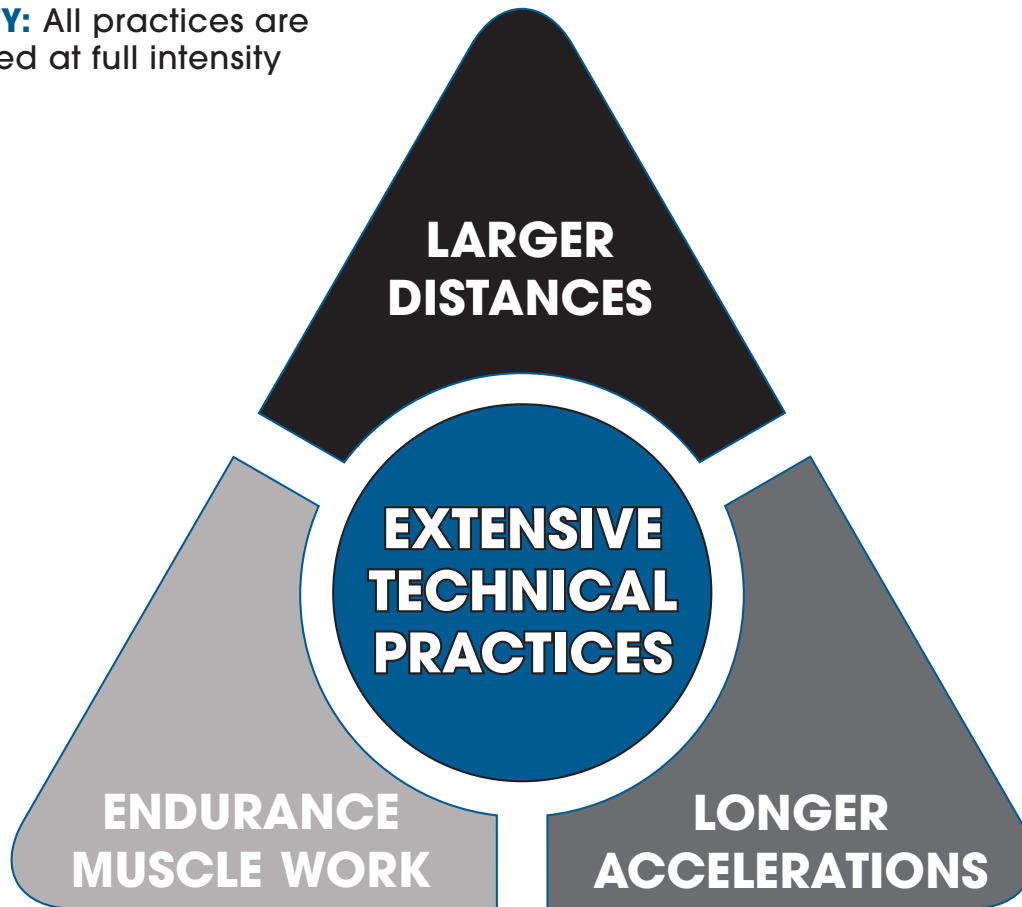
WEDNESDAY TRAINING SESSION (85-95 min)

Collective Team Principle Training and Speed Endurance Development:

1. Speed Endurance Warm-up (10-12 min)
2. Extensive Technical Practice (12-15 min)
3. Speed Endurance Conditioning Practice (5-15 min)
4. Large Sided Possession (10-15 min)
5. Large Sided Game in Large Area (10-50 min)

WEDNESDAY - 3 DAYS UNTIL MATCH (MD +4/-3): Extensive Technical Practices

INTENSITY: All practices are performed at full intensity



What are Extensive Technical Practices?

- Larger surface areas with longer passing ranges included within them (20+ metres).
- Larger surface areas to engage the key muscle groups required for the training session to follow.
- Longer acceleration and decelerations in larger spaces and training areas.
- They should be focused on keeping with the flow of the training day and working muscle groups on this particular day.

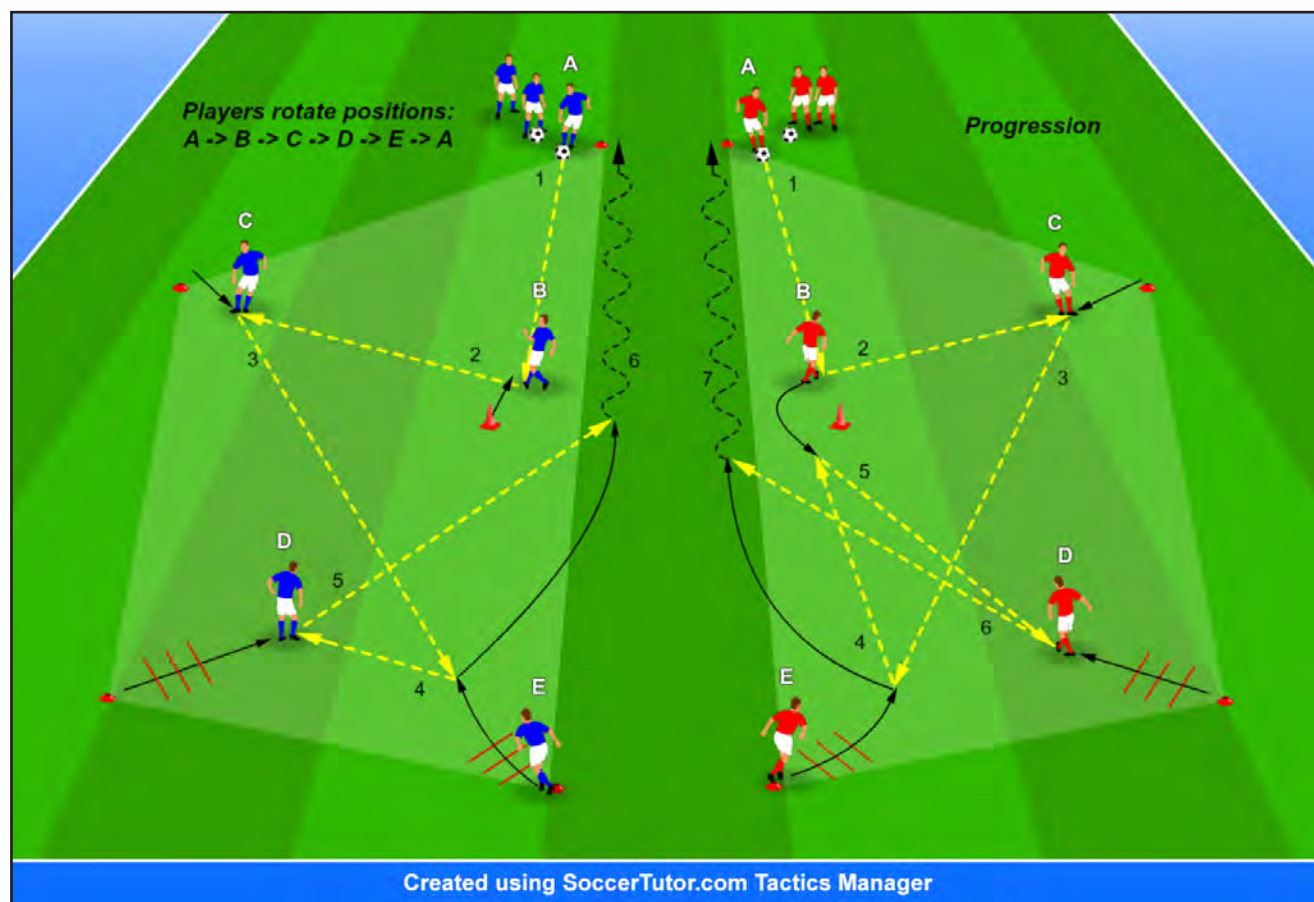
Why are they used on this day of the training week (MD +4/-3)?

- To prepare the players for the larger surface area type work developed through the course of the session.

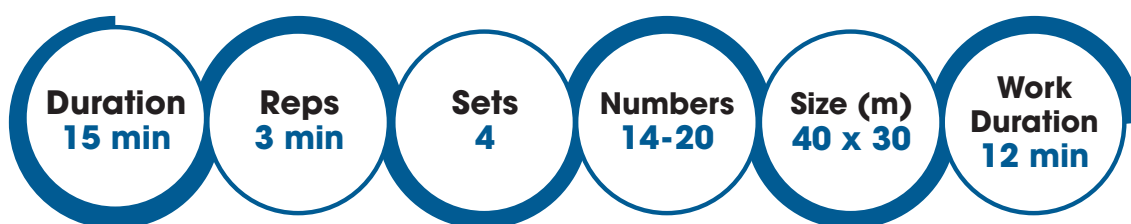
How does this help to maximise performance?

- As a way of preparing the players muscles used for covering larger distances, longer acceleration and deceleration efforts.
- Generally readying the body for the session ahead in the larger spaces (large sided games).

Extensive Technical 4: Fast Combination Play in a Pass & Move Drill with Final "Give & Go"



PRACTICE INFORMATION



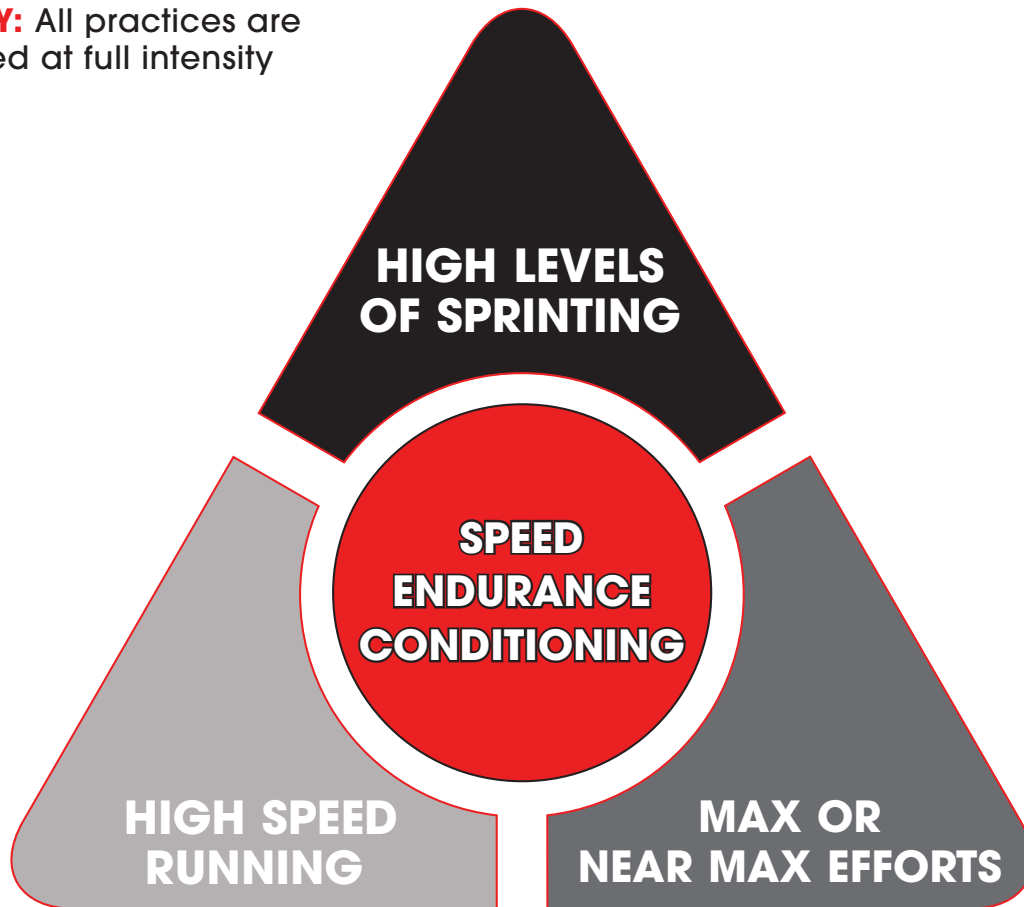
PRACTICE OBJECTIVES (2 BALLS): Ball speed (correct weight), timing of run, pass placement

Volume Metrics			Intensity Metrics		
	Practice Total	Per Min. of Work		Practice Total	Per Min. of Work
Total Distance (km)	1.41	0.12	Max Speed (m/s)	6.66	
High Speed Running (m)	57.82	4.82	Intensity (m/min)		94.39
Sprint Distance (m)	4.08	0.34	Power Score (w/kg)	7.98	
Work Ratio (%)	43.55		No. of Max Accels >4m ²	20.96	1.75
Power Plays (HiActs)	12.44	1.04	No. of Max Decels >4m ²	6.84	0.57

* The data shows the physical output per player based on research from elite level teams - see pages 81-83 for details

WEDNESDAY - 3 DAYS UNTIL MATCH (MD +4/-3): Speed Endurance Conditioning Practices

INTENSITY: All practices are performed at full intensity



What are Speed Endurance Conditioning Practices?

- These physical conditioning exercises are classed as 'speed endurance' as they expose players to high levels of sprinting and high-speed running metrics.
- They include larger surface areas with maximum or near maximum efforts to run or sprint.

Why are they used on this day of the training week (MD +4/-3)?

- To overload the key muscle groups required on a day that is far enough away from the match day to recover

but still elicit a stimulus in order to progress the players' capability to improve their conditioning.

How does this help to maximise performance?

- These high speed and sprint based practices are used on this day as a way of preparing and developing the players' muscles and capacity to perform this type of work in matches.

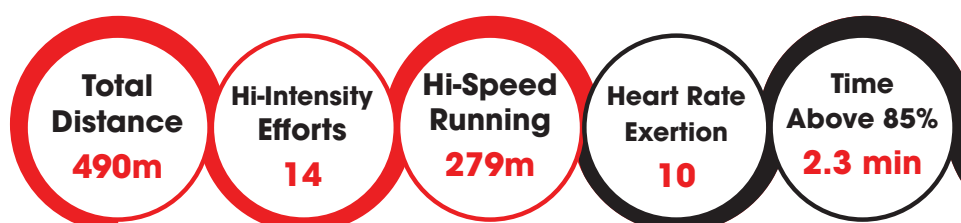
TECHNICAL / TACTICAL CONDITIONING

SPEED ENDURANCE: Based on 1 Set of 4 x reps (45s recovery)



Benefits of this Speed Endurance Practice?

- Balance between work and recovery
- High specific muscle overload
- Increase number of sets and reps as players become conditioned
- Allows players to improve playing intensity and recovery within game situations
- Improve players capacity to work and recover within the practice
- Technical and tactical focus
- Technical focus and quality is maintained throughout
- Improved motivation of players due to inclusion of technical element
- Ensure quality of movement and speed throughout!
- Direct link between the outcome of the practice/session and performance in a match



@adamowen1980

Speed Endurance Conditioning 6: Dribbling, Passing, and Finishing in a Sprinting Circuit



PRACTICE INFORMATION

Duration
12 min
Reps
4 x 60 m
Sets
2
Numbers
12+ (+GKs)
Size
As shown
Work
Duration
1.5 min
Explosive
HSR
Movements

OBJECTIVES: Physical stimulation - high-speed and sprints to develop the capacity to run fast

	Volume Metrics	Practice Total	Per Min. of Work
	Total Distance (km)	1.04	0.69
	High Speed Running (m)	304	202.67
	Sprint Distance (m)	50	33.33
	Work Ratio (%)	37.8	
	Power Plays (HiActs)	8	5.33

	Intensity Metrics	Practice Total	Per Min. of Work
	Max Speed (m/s)	7.44	
	Intensity (m/min)		89
	Power Score (w/kg)	4.45	
	No. of Max Accels >4m ²	2	1.33
	No. of Max Decels >4m ²	5	3.33

* The data shows the physical output per player based on research from elite level teams - see pages 81-83 for details

THURSDAY TRAINING DAY PRACTICES: **REACTION SPEED**

2 DAYS UNTIL MATCH (MD +5/-2)

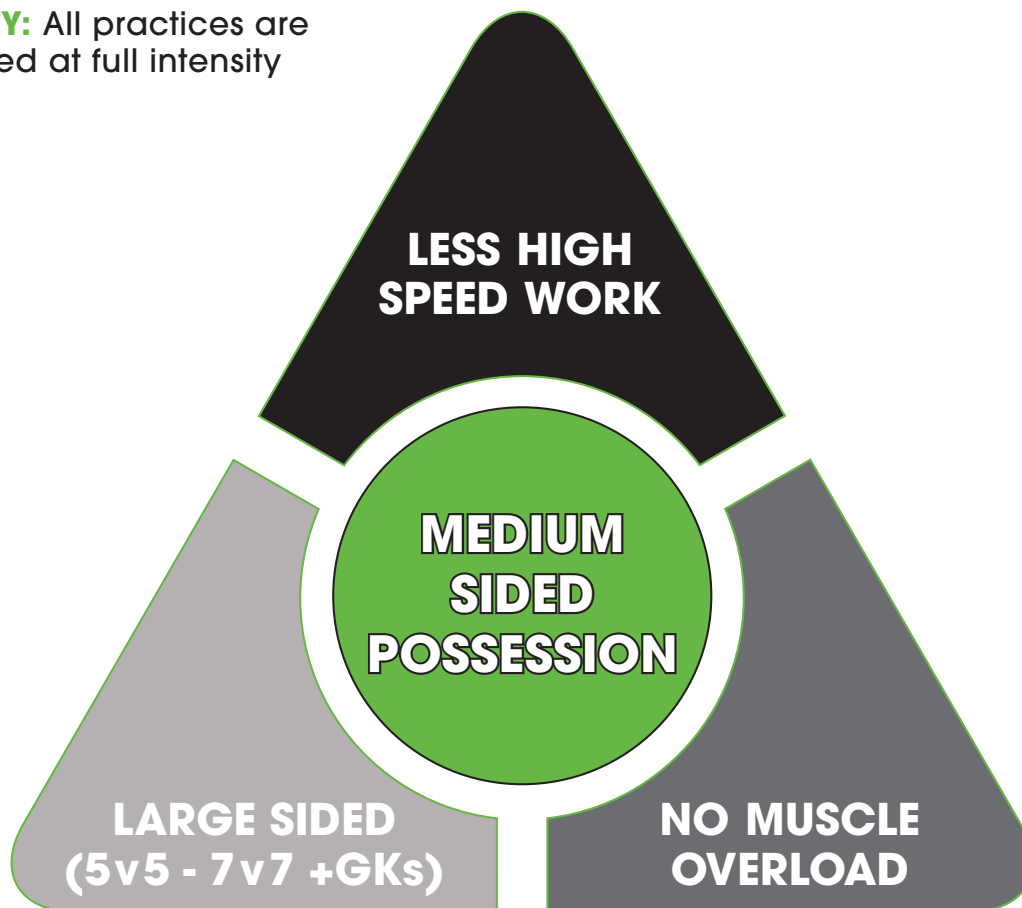
THURSDAY TRAINING SESSION (60-70 min)

Unit Principle Training and Reaction Speed Development:

1. Reaction Speed Warm-up (5-7 min)
2. Intensive Technical Practice (10-15 min)
3. Reaction Speed Conditioning Practice (5-15 min)
4. Medium Sided Possession (6-15 min)
5. Medium Sided Game (10-25 min)

THURSDAY - 2 DAYS UNTIL MATCH (MD +5/-2): Medium Sided Possession Practices

INTENSITY: All practices are performed at full intensity



What are Medium Sided Possession Practices?

- Medium sided possession games are defined as including 5 to 7 outfield players per side = 5v5 - 7v7 (+GKs).

Why are they used on this day of the training week (MD +5/-2)?

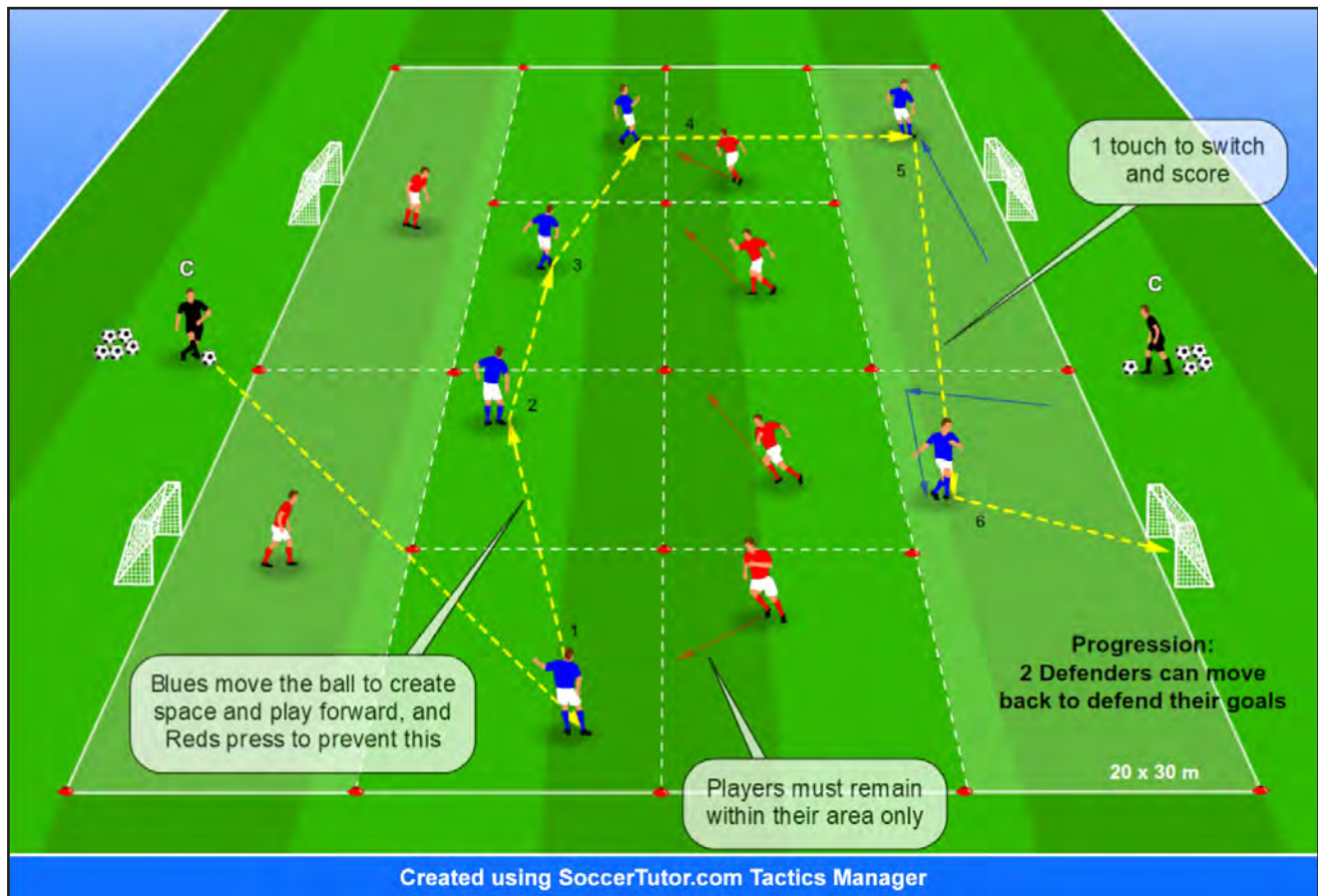
- They can be used in different periods of the training week but within this framework, they are focused around this training days with medium sized playing areas to train the unit principles (defence-midfield-attack).
- Medium sized areas are used to focus on specific areas of the pitch.

- Medium sided possession games are performed in this part of the training week to ensure specific muscle groups are not overloaded.

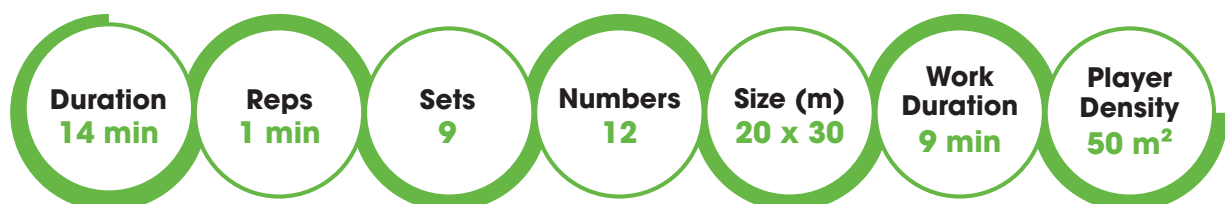
How does this help to maximise performance?

- Reduced high-speed running and sprint distance vs. larger game types.
- Expose players to reduced cardiovascular loads when compared to small sided possession games, due to a reduced intensity.

Medium Possession 6: Create Space and Play Through Press in a 6v6 Tactical Game



PRACTICE INFORMATION



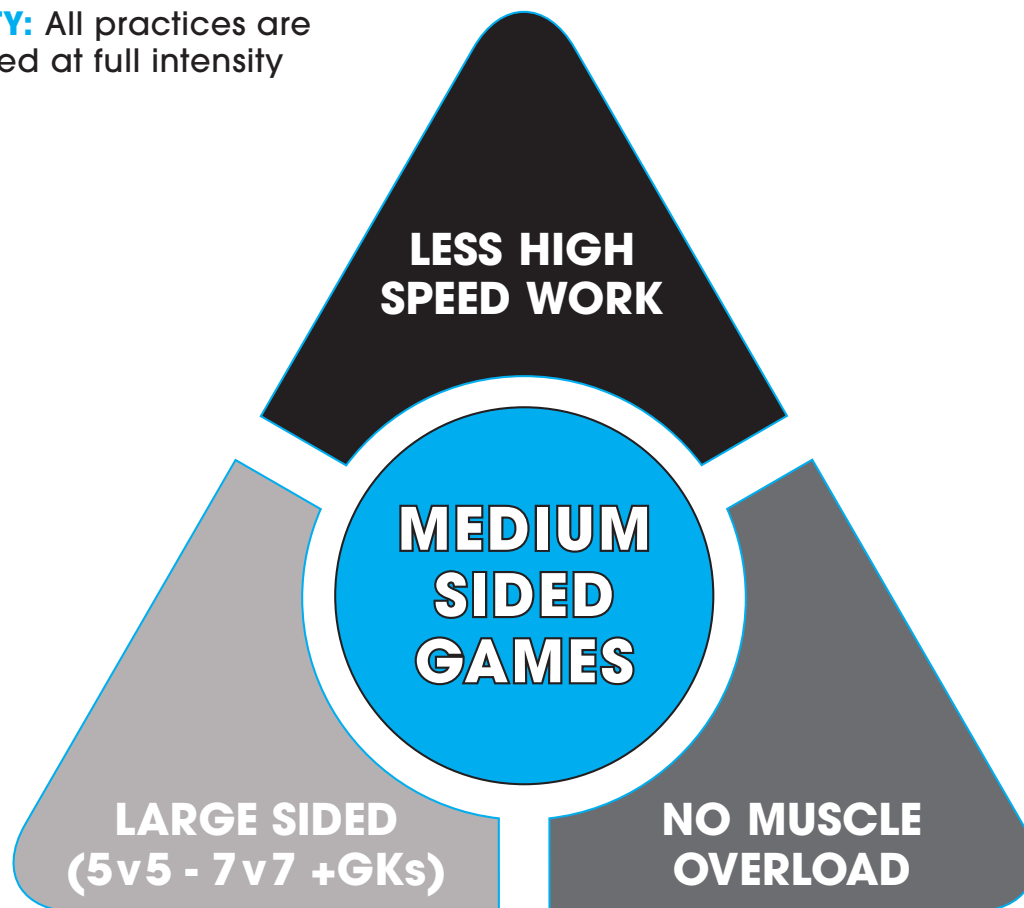
OBJECTIVE: Tactical possession - move ball, create space, play through the defensive line

Volume Metrics			Intensity Metrics		
	Practice Total	Per Min. of Work		Practice Total	Per Min. of Work
Total Distance (km)	1.061	0.12	Max Speed (m/s)	5.15	
High Speed Running (m)	3.98	0.44	Intensity (m/min)		51.99
Sprint Distance (m)	-	-	Power Score (w/kg)	4.51	
HML Distance (m)	152.72	16.97	No. of Max Accels >4m ²	14.75	1.64
Power Plays (HiActs)	1.33	0.15	No. of Max Decels >4m ²	16	1.78

* The data shows the physical output per player based on research from elite level teams - see pages 81-83 for details

THURSDAY - 2 DAYS UNTIL MATCH (MD +5/-2): Medium Sided Games

INTENSITY: All practices are performed at full intensity



What are Medium Sided Games?

- Medium sided games (MSGs) are defined as including 5 to 7 outfield players per side = 5v5 - 7v7 (+GKs).

Why are Medium Sided Games used at this point of the training week (MD +5/-2)?

- They can be used in different periods of the training week but within this framework, they are focused around this training days with medium sized playing areas to train the unit principles (defence-midfield-attack).
- Medium sized areas are used to focus on specific areas of the pitch.

- Medium sided games are performed in this part of the training week to ensure specific muscle groups are not overloaded.

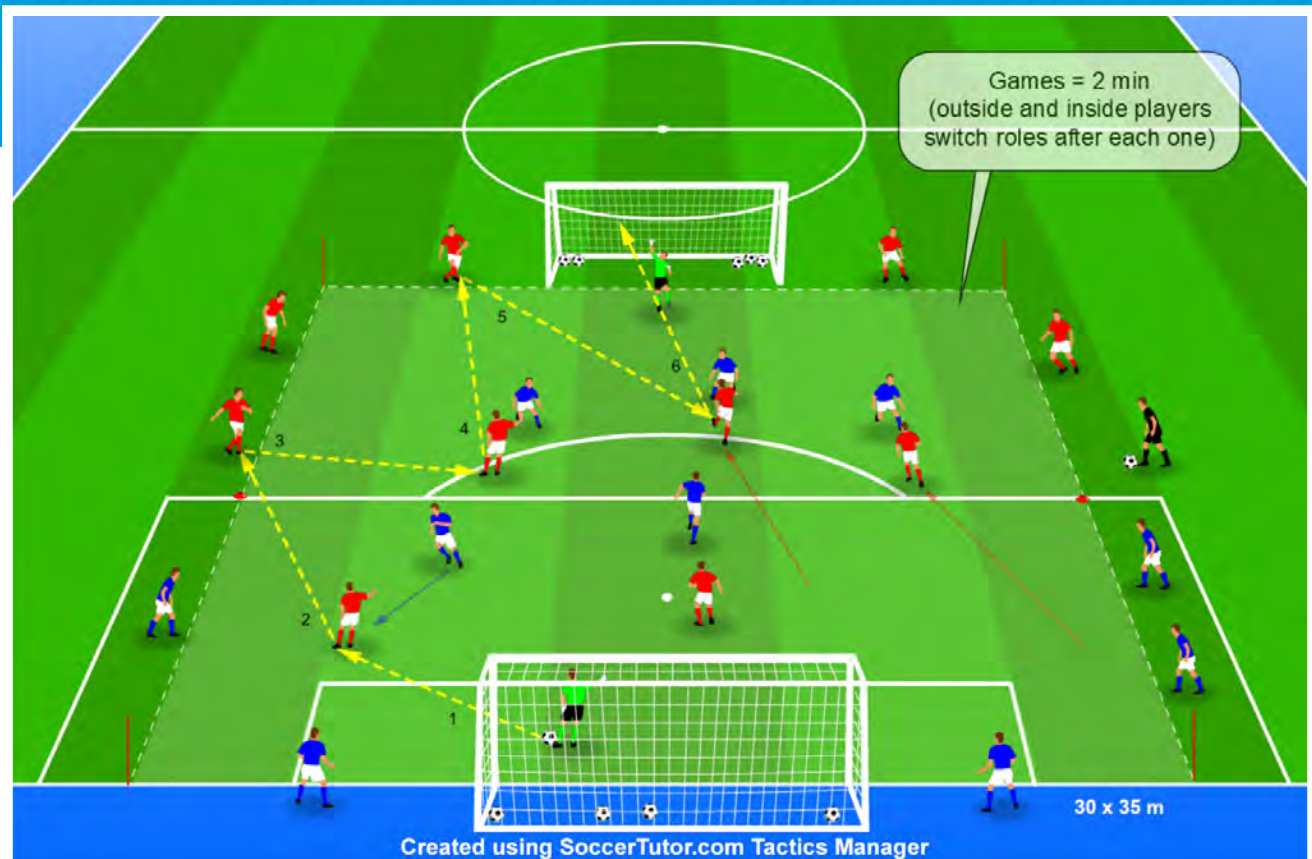
How does this help to maximise performance?

- These game types result in reduced high-speed running and sprint distance vs. larger game types.
- But expose players to reduced cardiovascular loads when compared to SSGs, due to a reduced intensity.

MEDIUM SIDED GAMES

5 (+5) v 5 (+5) + GKs - 30 x 35 m

*Data below based on 4 x 2 Minute Playing Sets per Team



↑ Football Specific Training

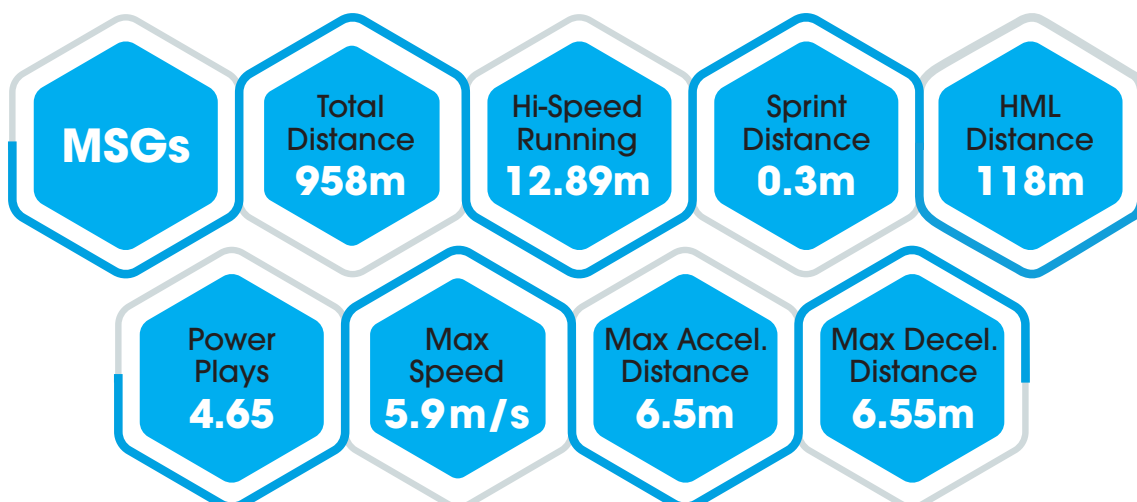
↑ Speed of Play

↑ Technical Exposure

↑ Cardiovascular Load

↑ Goal Scoring Opportunities

↑ 1 v 1 Attacking / Defending Situations

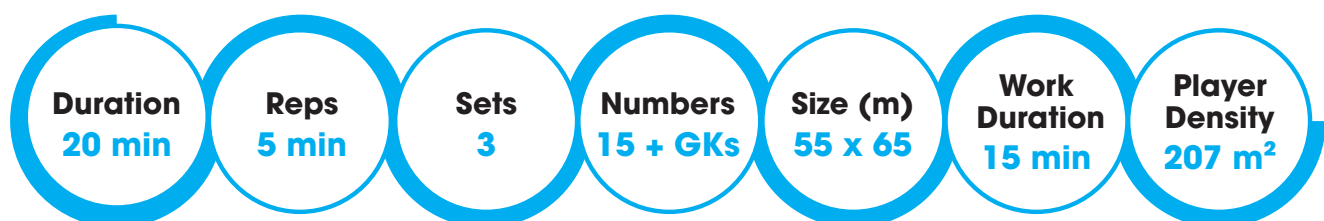


@adamowen1980

Medium Sided Game 1: Counter Attacking in a Three Team 5v5v5 (+GKs) Game



PRACTICE INFORMATION



OBJECTIVE: Transition based three team training game with focus on counter attacking

Volume Metrics			Intensity Metrics		
	Practice Total	Per Min. of Work		Practice Total	Per Min. of Work
Total Distance (km)	1.625	0.1	Max Speed (m/s)	6.8	
High Speed Running (m)	73.79	4.92	Intensity (m/min)		79.98
Sprint Distance (m)	4.43	0.3	Power Score (w/kg)	6.86	
Work Ratio (%)	31.36		Max Accel. Distance (m)	21.58	1.44
Power Plays (HiActs)	12.94	0.86	Max Decel. Distance (m)	30.05	2

* The data shows the physical output per player based on research from elite level teams - see pages 81-83 for details

FRIDAY TRAINING DAY PRACTICES:

PRE-MATCH ACTIVATION

1 DAY UNTIL MATCH (MD +6/-1)

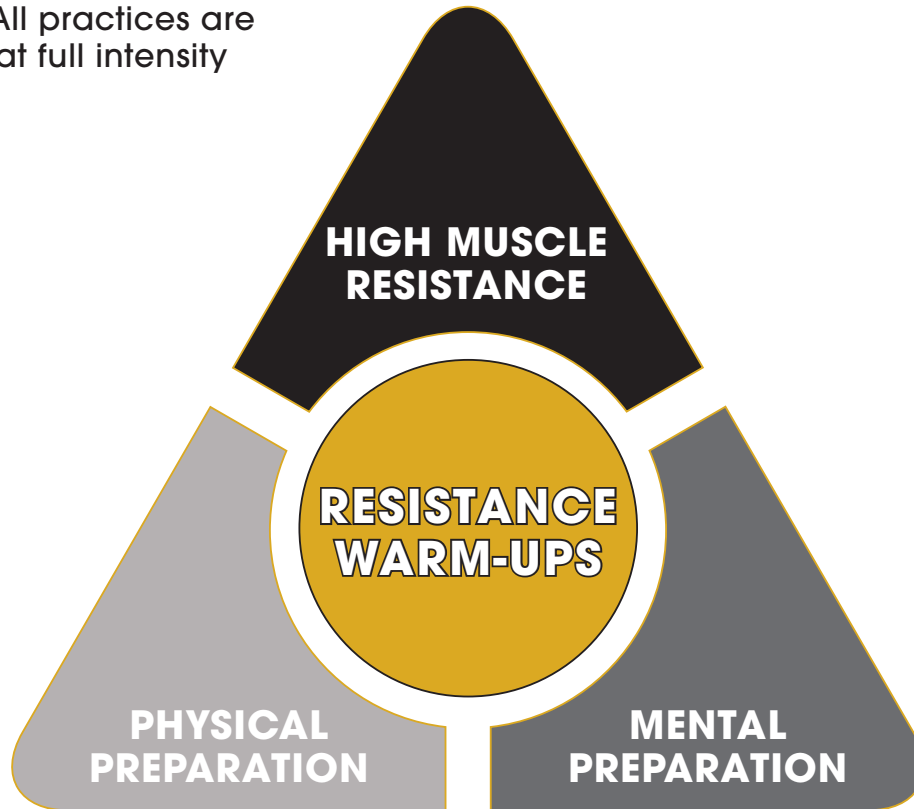
FRIDAY TRAINING SESSION (45-60 min)

Pre-Match Activation Training Day:

1. Resistance Warm-up (10-12 min)
2. Reaction Speed Conditioning Practice (5-15 min)
3. Large Sided Game in Small/Medium Area (10-50 min)

FRIDAY - 1 DAY UNTIL MATCH (MD +6/-1): Resistance Warm-Up Practices

INTENSITY: All practices are performed at full intensity



What are Resistance Warm-ups?

- Include many stop and start actions, and directional changes.
- Include many lower level accelerations and decelerations in tight spaces.
- Activate the muscle groups for the explosive maximum accelerations and decelerations later in the session.
- Provide more resistance to the working muscles through explosive actions in small spaces.

Why are they used on this day of the training week (MD +3/-4)?

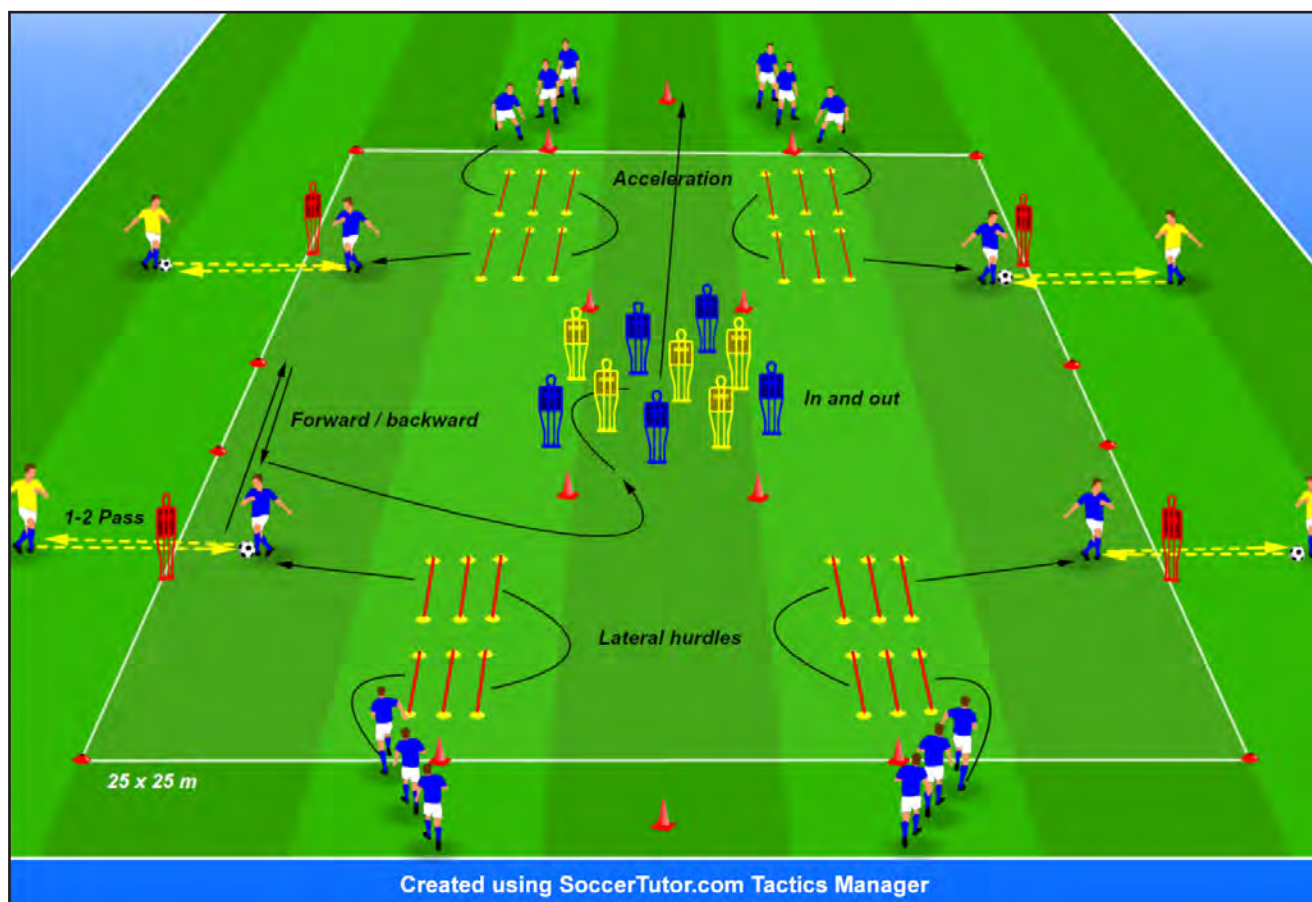
- To prepare the players for the smaller surface area type work developed through the course of the session.

How does this help to maximise performance?

- Resistance warm-ups are used on this day as a way of preparing the players muscles used for changing directions, acceleration and deceleration efforts.
- Resistance warm-ups also generally ready the body for the session ahead (small sided games).

NOTE: There are 2 more Resistance Warm-up Practices you can use from the Tuesday Training Day (pages 97-98) - this section starts from Practice No.3...

Resistance Warm-up 3: Lateral Hurdles, One-Two, In-and-Out Movements + Sprint Circuit



PRACTICE INFORMATION

Duration
10 min

Reps
2 min

Sets
4

Numbers
12-20

Size (m)
25x25

Work Duration
5 min

OBJECTIVE: Physically and mentally prepare the players for the training session (with the ball)

Volume Metrics			Intensity Metrics		
	Practice Total	Per Min. of Work		Practice Total	Per Min. of Work
Total Distance (km)	0.91	0.18	Max Speed (m/s)	4.78	
High Speed Running (m)	0.18	0.036	Intensity (m/min)		111.32
Sprint Distance (m)	-	-	Power Score (w/kg)	5.41	
Work Ratio (%)	27.3		No. of Max Accels >4m ²	12.59	2.52
Power Plays (HiActs)	1.89	0.38	No. of Max Decels >4m ²	9.61	1.92

* The data shows the physical output per player based on research from elite level teams - see pages 81-83 for details

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